

The Zero Trust Credit Union

Securing Member Information Without Sacrificing Member Experience

Executive Summary

Credit unions have invested heavily in cybersecurity over the past decade. Multi-factor authentication, endpoint protection, identity management, network segmentation, and Zero Trust initiatives have become foundational components of modern security programs.

Yet despite these investments, sensitive member information continues to be exposed through ransomware attacks, insider threats, third-party relationships, and unsecured collaboration processes.

The challenge is not that credit unions have failed to adopt Zero Trust principles. The challenge is that most Zero Trust initiatives stop at identity, devices, and networks while leaving a critical element largely unaddressed: data access.

As credit unions modernize member services, expand digital lending operations, collaborate with external partners, and support hybrid work environments, data is constantly moving between employees, members, service providers, auditors, regulators, attorneys, mortgage brokers, and business partners.

This creates a growing gap between traditional Zero Trust controls and the actual ways sensitive information is accessed, shared, and used.

This paper introduces the concept of the Zero Trust Credit Union and presents a framework for extending Zero Trust principles directly to data access and collaboration.

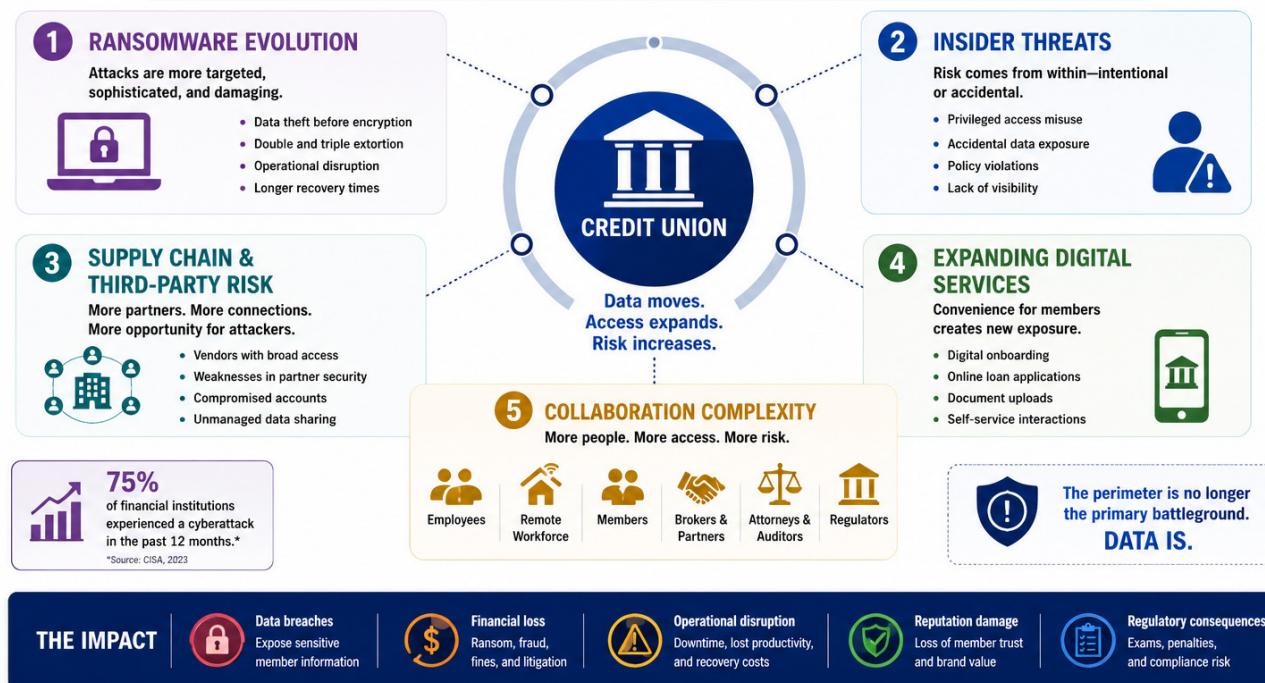
The result is a security model that enables credit unions to improve member experiences while maintaining control over their most valuable asset: member information.

The New Threat Landscape

THE CREDIT UNION THREAT LANDSCAPE

EVOLVING THREATS. EXPANDING ATTACK SURFACE. HIGHER STAKES.

Cybercriminals target what matters most: **SENSITIVE MEMBER INFORMATION**



To protect members and the mission, credit unions must extend **Zero Trust** beyond identity and network security to the **access and movement of data**.

The threat environment facing financial institutions has changed dramatically.

Traditional security architectures were designed to protect internal networks and trusted users. Modern attacks target data itself.

Cybercriminals no longer need to breach an entire network to achieve their objectives. They only need access to sensitive information.

Several trends are accelerating this risk.

Ransomware Evolution

Ransomware attacks have evolved from simple encryption attacks into sophisticated data theft and extortion campaigns.

Attackers increasingly focus on:

- Sensitive member records
- Financial information
- Loan documentation
- Internal communications

- Regulatory documents

Even organizations with strong perimeter defenses can suffer significant damage if attackers gain access to critical data repositories.

Insider Threats

Not all threats originate from outside the organization.

Employees, contractors, consultants, and third-party partners often require legitimate access to sensitive information.

Whether intentional or accidental, excessive permissions and uncontrolled data sharing can expose member information without triggering traditional security controls.

Supply Chain and Third-Party Risk

Credit unions increasingly rely on external partners including:

- Mortgage brokers
- Attorneys
- Auditors
- Collection agencies
- Consultants
- Technology providers

Each relationship introduces new data-sharing requirements and expands the attack surface.

Expanding Digital Services

Members expect frictionless digital experiences.

Loan applications, document submissions, account servicing, and collaboration increasingly occur through digital channels.

While these services improve convenience, they also create additional pathways through which sensitive information is exchanged.

The result is a simple reality:

Data is moving farther, faster, and through more participants than ever before.

The Evolution of Zero Trust

To understand the next stage of cybersecurity, it is helpful to examine how security models have evolved.

Castle and Moat Security

Historically, organizations operated under a perimeter-based model.

Users inside the network were trusted.

Users outside the network were not.

Security investments focused on:

- Firewalls
- VPNs
- Network boundaries

This approach worked when users, applications, and data remained within clearly defined environments.

Identity-Centric Security

Cloud computing, remote work, and mobile devices challenged traditional assumptions.

Organizations responded by shifting trust decisions toward identity.

Technologies such as:

- Multi-factor authentication
- Identity governance
- Single sign-on
- Conditional access

became central components of modern security programs.

Zero Trust Security

The Zero Trust model emerged to address the limitations of perimeter-based security.

Its guiding principle is simple:

Never trust. Always verify.

Users, devices, applications, and networks are continuously validated before access is granted.

The Evolution of Zero Trust

Era	Security Model	Core Assumption	Primary Controls	Strengths	Limitations
1990s–2000s	Castle-and-Moat Security	Users inside the network can be trusted.	Firewalls, VPNs, perimeter defenses	Effective when users, applications, and data remained inside the corporate network.	Once attackers breach the perimeter, they often gain broad access.
2000s–2010s	Perimeter Security 2.0	Strengthen the perimeter to keep threats out.	IDS/IPS, web gateways, network monitoring, VPNs	Improved visibility and threat detection.	Data, applications, and users increasingly moved beyond the perimeter.
2010s	Identity-Centric Security	Trust should be based on verified identity rather than network location.	MFA, SSO, IAM, Conditional Access	Stronger user verification and reduced credential risk.	Verifies who the user is but provides limited control over what happens to data after access is granted.
2020s	Zero Trust Security	Never trust. Always verify.	MFA, device trust, endpoint security, micro-segmentation, continuous validation	Reduces implicit trust and improves security posture.	Most implementations focus on users, devices, and networks rather than data access and collaboration.
Emerging	Zero Trust Data Access	Access to data must be continuously governed regardless of user, device, or location.	Granular permissions, context-aware access, activity monitoring, audit trails, data governance controls	Protects information throughout its lifecycle while enabling secure collaboration.	Requires organizations to extend Zero Trust principles beyond traditional security controls.

The Missing Layer: Zero Trust Data Access

The next evolution of Zero Trust is extending protection directly to data.

Consider a common scenario.

An employee successfully authenticates through MFA.

Their device is compliant.

Their network connection is approved.

Traditional Zero Trust controls have done their job.

***“Perimeter Security protected networks.
Identity Security protected users.
Zero Trust protected access.
Zero Trust Data Access protects information”***

But several important questions remain:

- What data should this user access?
- How long should access remain available?
- Can the information be downloaded?
- Can it be shared externally?
- Can it be copied to unmanaged devices?

- Can activity be audited?

Identity verification alone does not answer these questions.

The challenge is that access decisions often end at authentication.

Data controls frequently remain static, broad, and difficult to manage.
This creates a gap between verifying users and protecting information.

Defining Zero Trust Data Access

Zero Trust Data Access extends Zero Trust principles directly to information itself.
Under this model:

- Access is granted based on context
- Permissions are granular
- Activity is continuously monitored
- Data remains under organizational control
- Sharing is governed by policy
- Auditability is maintained throughout the lifecycle of access

Rather than assuming trust after authentication, organizations continuously validate how information is accessed and used.

This approach aligns security controls with the asset attackers ultimately seek: data.
Under this model:

- Access is granted based on context
- Permissions are granular
- Activity is continuously monitored
- Data remains under organizational control
- Sharing is governed by policy
- Auditability is maintained throughout the lifecycle of access

Rather than assuming trust after authentication, organizations continuously validate how information is accessed and used.

This approach aligns security controls with the asset attackers ultimately seek: data.

THE EVOLUTION OF SECURITY AND ZERO TRUST

From protecting the network to protecting what matters most: information.



FOCUS AREA	NETWORK	NETWORK	IDENTITY	IDENTITY + DEVICE	DATA
TRUST MODEL	TRUST INSIDE	VERIFY PERIMETER	VERIFY USER	CONTINUOUS VERIFICATION	CONTINUOUS DATA GOVERNANCE
PROTECTED ASSET	NETWORK	NETWORK	USER ACCOUNTS	SYSTEMS & APPLICATIONS	INFORMATION
KEY QUESTION	"Are you inside?"	"Can you get in?"	"Who are you?"	"Should you have access?"	"How is the data being used?"

THE NEXT EVOLUTION: Zero Trust Data Access extends Zero Trust principles to the data itself—ensuring sensitive information is accessed, shared, and used securely across every interaction.

Credit Union Collaboration Risks

Many of the highest-risk processes within credit unions involve collaboration.

Mortgage Workflows

Mortgage transactions often involve:

- Applicants
- Loan officers
- Underwriters
- Attorneys
- Appraisers
- Brokers

Sensitive information moves repeatedly among internal and external participants.

Email attachments and unmanaged file-sharing methods create visibility and governance challenges.

Lending Operations

Commercial and consumer lending frequently require collaboration across departments and external stakeholders.

Without consistent access controls, information can be duplicated, downloaded, or distributed beyond intended recipients.

Board Communications

Board packets often contain highly sensitive information.

Financial reports, strategic plans, acquisition discussions, and regulatory matters require secure distribution and controlled access.

Member Interactions

Members increasingly expect digital self-service experiences.

Uploading, downloading, reviewing, and exchanging sensitive documents must be both convenient and secure.

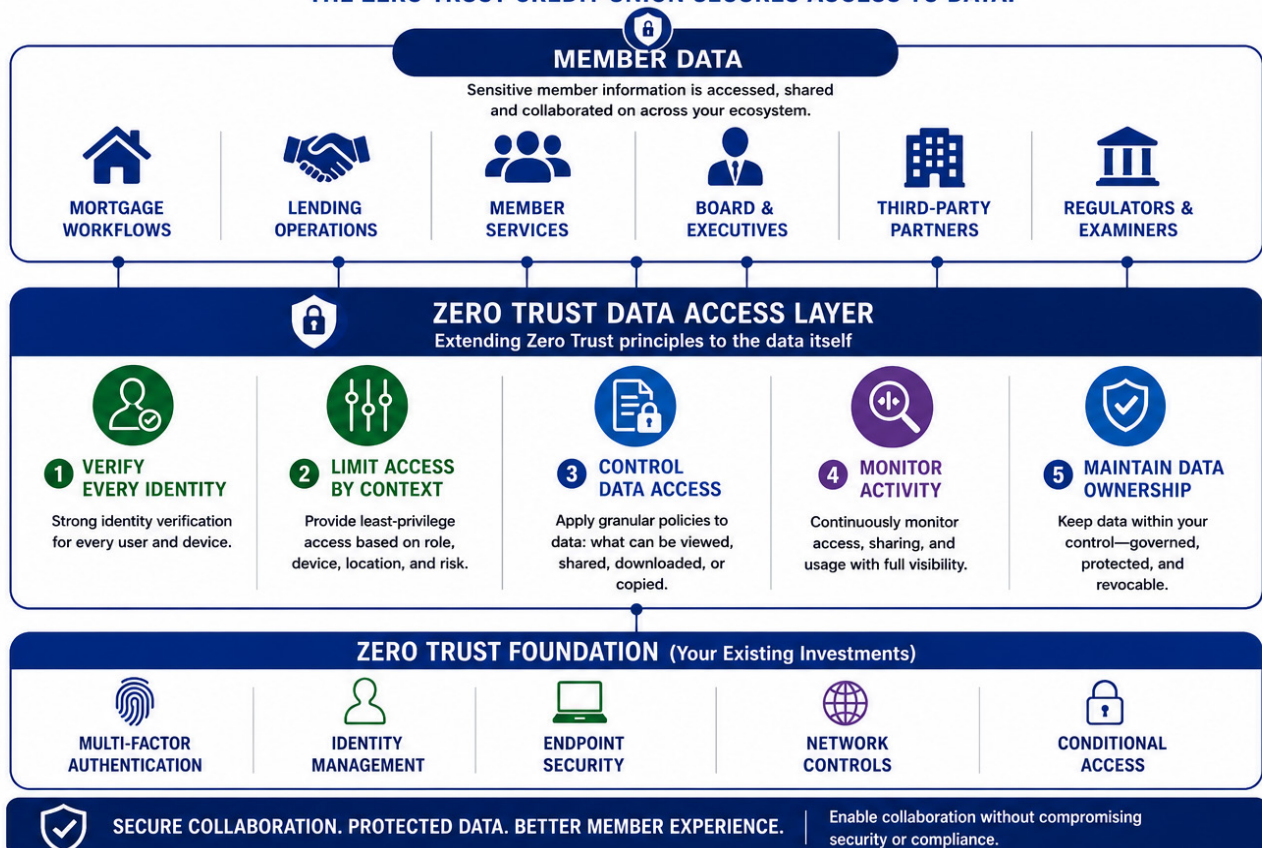
The challenge is not simply protecting information from attackers.

It is ensuring that every participant receives the appropriate level of access without creating unnecessary risk.

The Zero Trust Credit Union Framework

THE ZERO TRUST CREDIT UNION FRAMEWORK™

TRADITIONAL ZERO TRUST SECURES ACCESS TO SYSTEMS.
THE ZERO TRUST CREDIT UNION SECURES ACCESS TO DATA.



The Zero Trust Credit Union extends traditional Zero Trust principles beyond identity and network controls.

The framework consists of five pillars.

Pillar 1: Verify Every Identity

Authenticate every user, device, and participant.

Pillar 2: Limit Access by Context

Provide only the access necessary to complete the required task.

Pillar 3: Control Data Access

Apply policies directly to information.

Determine:

- Who can access data
- What actions are permitted
- How long access remains available

Pillar 4: Continuously Monitor Activity

Maintain visibility into how information is accessed, shared, and used.

Generate auditable records that support compliance and risk management.

Pillar 5: Keep Data Under Organizational Control

Reduce unnecessary data replication and movement.

Enable collaboration while preserving governance and oversight.

Together, these pillars create a practical framework for modern financial institutions.

A Roadmap for Implementation

Most credit unions do not need to replace existing security investments.

Instead, they can extend current Zero Trust initiatives by focusing on data access and

Phase 1: Assess

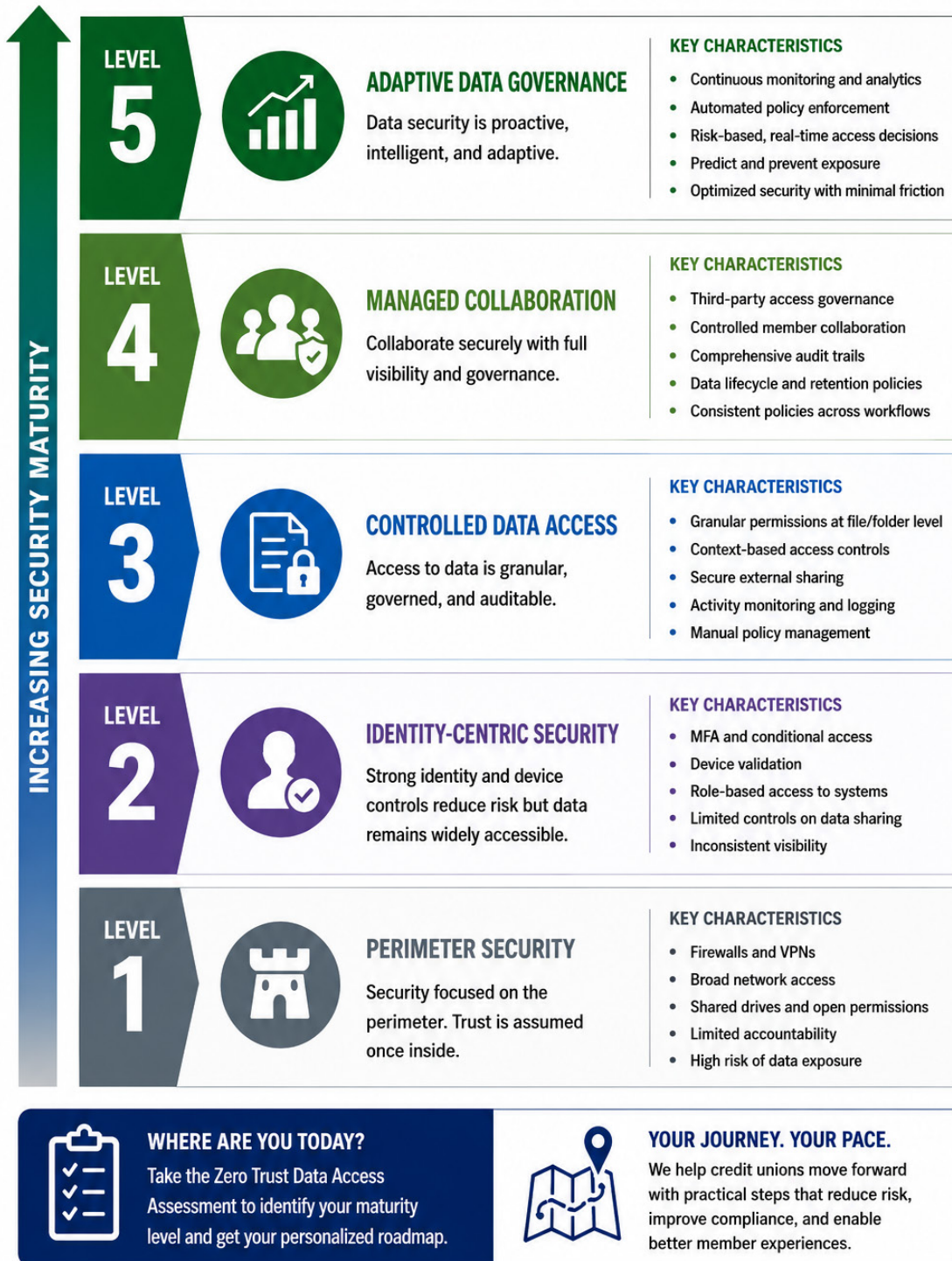
Identify high-risk collaboration workflows.

Examples include:

- Mortgage processing

ZERO TRUST DATA ACCESS MATURITY MODEL™

A Five-Level Journey to Protecting Member Information



- Lending operations
- Board communications
- Third-party collaboration

Phase 2: Classify

Determine which information assets require enhanced controls.

Phase 3: Govern

Define policies governing access, sharing, and auditing.

Phase 4: Implement

Deploy technologies that enforce Zero Trust Data Access principles.

Phase 5: Measure

Track:

- Reduction in unsecured sharing
- Improved auditability
- Compliance outcomes
- User adoption
- Operational efficiency

Over time, these improvements create a stronger security posture while improving member experiences and operational effectiveness.

Conclusion

The cybersecurity landscape continues to evolve.

Identity, endpoints, and networks remain essential components of a Zero Trust strategy, but they are no longer sufficient on their own.

As credit unions accelerate digital transformation, collaboration has become a primary pathway through which sensitive information is accessed and shared.

Protecting data requires extending Zero Trust principles beyond authentication and network controls to the information itself.

The Zero Trust Credit Union framework provides a practical approach for securing member information, enabling collaboration, supporting compliance, and reducing organizational risk.

The future of Zero Trust is not simply verifying who accesses systems.

It is continuously governing how critical information is accessed, shared, and protected.